

PROJECT REPORT

AI Job Tracker System (Pre-Login Version)

◆ 1. INTRODUCTION

The **AI Job Tracker** is a web-based application designed to help users track their job applications efficiently.

It combines:

- **Frontend (React)** for UI
- **Google Sheets** as database
- **n8n automation** as backend

👉 The system allows users to:

- Add job applications
 - Track status (Interview, Offer, Rejected)
 - View insights through dashboard
 - Automatically manage follow-up dates
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◆ 2. OBJECTIVE

To build a smart job tracking system that:

- ✓ Stores job applications
 - ✓ Tracks progress
 - ✓ Automates follow-up dates
 - ✓ Provides real-time dashboard
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◆ 3. TECHNOLOGIES USED

Frontend

- React
- JavaScript

- CSS (inline styling)
 - Recharts (for graphs)
 - Lucide React (icons)
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● Backend (Automation)

- n8n
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● Database

- Google Sheets
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◆ 4. SYSTEM ARCHITECTURE

React Frontend



Webhook API (n8n)



Google Sheets Database



Data fetched back to React UI

◆ 5. FEATURES IMPLEMENTED

● 5.1 Add Job Application

User enters:

Company

Role

Status

👉 On clicking **Add Job**:

- Data sent to n8n webhook
 - Stored in Google Sheets
 - UI updates instantly (Optimistic UI)
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● 5.2 View Jobs Dashboard

- All jobs displayed as **cards**
 - Each card shows:
 - Company
 - Role
 - Status
 - Date
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● 5.3 Filter System

All | Interview | Offer | Rejected

👉 Allows quick categorization of applications

● 5.4 Search Functionality

- Search by company name
 - Real-time filtering
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● 5.5 Edit Job Status

- Click on job card → open modal
 - Update status
 - Changes saved via n8n
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● 5.6 Automatic Date System

👉 Instead of asking user:

Next Action Date = Today + 3 days

- ✓ Reduces user effort
 - ✓ Improves usability
-

● 5.7 Optimistic UI (Advanced Feature)

User action → UI updates instantly → backend updates later

- ✓ Removes lag
 - ✓ Improves experience
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● 5.8 Loading Indicators

- “Adding...” button state
 - Loader while fetching jobs
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● 5.9 Stats Dashboard

Displays:

Total Jobs

Interviews

Offers

Rejected

● 5.10 Analytics Chart

Using Recharts:

- Bar chart visualization
 - Shows job distribution
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● 5.11 Responsive UI

- Works on different screen sizes
- Flexible layout using grid

◆ 6. n8n WORKFLOWS

● Workflow 1 — Add Job

Webhook → Append Row (Google Sheets)

Data stored:

Company

Role

Status

Date (auto generated)

● Workflow 2 — Update Job

Webhook → Update Row

👉 Updates status dynamically

● Workflow 3 — (Optional earlier)

Schedule → Read → Gmail → Update

👉 Used for automation (optional feature)

◆ 7. GOOGLE SHEET STRUCTURE

Column A → Company

Column B → Role

Column C → Status

Column D → Notes

Column E → Next Action Date

◆ 8. PROBLEMS FACED & SOLUTIONS

● Problem 1: UI Lag

Cause → waiting for backend response

✓ Solution:

Optimistic UI

● Problem 2: Data not updating instantly

Cause → Google Sheets delay

✓ Solution:

setTimeout + delayed fetch

● Problem 3: Date mismatch

Cause → timezone difference

✓ Solution:

Use UTC-based date logic

● Problem 4: Date disappearing after refresh

Cause → not stored in sheet

✓ Solution:

Add date field in n8n

◆ 9. OUTPUT

✓ Fully working job tracker

✓ Real-time UI

✓ Data stored in cloud (Sheets)

✓ Interactive dashboard

✓ Analytics visualization

◆ 10. LIMITATIONS

- ✗ No login system
- ✗ Single user only
- ✗ Not secure authentication
- ✗ Depends on Google Sheets speed

◆ 11. FUTURE SCOPE

- ✓ Multi-user login system
- ✓ AI-based email detection
- ✓ Notifications / reminders
- ✓ Backend (Node.js + DB)
- ✓ Deployment (Vercel)

◆ 12. CONCLUSION

The project successfully demonstrates:

- ✓ Integration of frontend + automation + cloud DB
 - ✓ Real-time user interaction
 - ✓ Efficient job tracking system
- 👉 It serves as a **strong foundation for a SaaS product.**

🔲 FINAL NOTE

This is not just a project anymore: You built a mini product system 🚀